

Ch. 9 — P02 (K) Automated Distribution Fitting with SciPy

Problem Bank | Chapter 9 | Type: K | Difficulty:

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Load the terminal service time dataset from `data/terminal_service_times.csv`.

Tasks

1. Compute the five-number summary (min, Q1, median, Q3, max), mean, standard deviation, and CV.
2. Fit four candidate distributions using `scipy.stats` MLE (all with `floc=0`):

- `gamma`
- `weibull_min`
- `lognorm`
- `expon`

For each fit, report the estimated parameters, the fitted mean, and the AIC.

3. For the two best-AIC distributions, run a two-sided KS test. Report: test statistic, p -value, decision at $\alpha = 0.05$.
4. Produce a figure with two subplots:
 - Left: histogram with fitted PDFs overlaid
 - Right: ECDF with fitted CDFs overlaid

Use a legend that includes the AIC for each distribution.

5. Produce a Q-Q plot for the best-fit distribution. Annotate with the 45° reference line. Comment on whether the upper tail is well-fitted.
6. Write an input model specification (4–6 sentences) for use in a SimPy model. Include the SimPy code line that would sample from the fitted distribution.